

Solution Requirements

Date	01 October 2026
Team ID	SWTID-2026-9629
Project Name	PocketSmart AI: Your Smart Budget & Recommendation Assistant
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

The requirements below define what PocketSmart AI does and how well it performs. Targets in the non-functional table are design targets for the project.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users register (/register), log in (/login) and log out (/logout). On successful login the system issues a secure JWT through /token and starts a session; user credentials are stored securely.	High
2	Authorization levels	Single role: registered user. Planner pages, /history and session routes are protected and need an authenticated active user; each user can see only their own history and session data. Home and testimonials pages remain public.	High
3	External interfaces	Google Gemini 1.5 Flash Pro API (text and image input) for recommendation generation; product / service sources Amazon, Flipkart, IKEA, Swiggy, Zomato and OYO, linked through mock / simulated API calls.	High
4	Transactions processing	Each request flows: validated form input, prompt built in gemini_utils.py, Gemini call, structured JSON response, rendering on the recommendation page, and saving of the query and result to history. No payments are handled.	High
5	Reporting	User dashboard with recent recommendations, saved queries and insights; /history page listing past queries and results; /recommendations-details for category-wise AI recommendations.	Medium
6	Business rules, etc.	Recommendations must stay within the entered budget; budget is split by room / category (Home) and proportionally across catering, decoration and entertainment (Party); suggestions adapt to event type and occasion; outfit image is optional for Jewelry; fallback recommendations are shown when the AI result is insufficient.	High
7	Compliance to laws or regulations	Protect personal data and uploaded outfit images in line with applicable data-privacy norms (for example India's DPDP Act, 2023); API keys kept in a .env file and never committed; platform content used only as links / mock data.	Medium
8	Other	Responsive HTML/CSS UI, input validation for edge cases, CORS setup for frontend communication and graceful error messages.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Pages should load quickly; AI recommendation requests should return within a reasonable time even though they depend on Gemini.	Page loads in < 2 seconds; AI recommendation response in < 10 seconds
2	Scalability	Modular, stateless FastAPI endpoints so the app can handle more users and can be containerised and scaled horizontally later.	Supports 100 concurrent users in testing
3	Security & Data Privacy	JWT-based authentication, secure credential storage, API key in .env, CORS configuration and validation of all user inputs and uploads.	No secrets in source control; all planner routes reject unauthenticated requests
4	Reliability & Availability	Fallback / default recommendations and error handling when the AI returns insufficient results or the API is unavailable.	99% uptime during demo period; 0 unhandled server errors in test runs
5	Usability & Accessibility	Clean, card-based, responsive interface with clear forms for each planner and readable recommendation layout.	Works on desktop and mobile widths; aligned with WCAG 2.1 AA colour-contrast guidance
6	Other	Maintainability: separation into routes, services (gemini_utils.py) and models. Portability: runs on any OS with Python 3 and pip.	Setup completed in < 10 minutes using README